

Dimensions and their Measurement

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Dimension

A numerical value expressed in appropriate units of measure, indicated on drawings with lines, symbols, and notes to define the size or geometric characteristics of a part or feature.

Tolerance

The total amount by which a specific dimension is permitted to vary. It's the difference between the maximum and minimum limits of a dimension.

Bilateral Tolerance

Variation is permitted in both positive and negative directions from the nominal dimension.

Unilateral Tolerance

Variation is permitted in only one direction, either positive or negative, from the nominal dimension.

Limit Dimensions

Alternative method specifying maximum and minimum allowable dimensions directly.

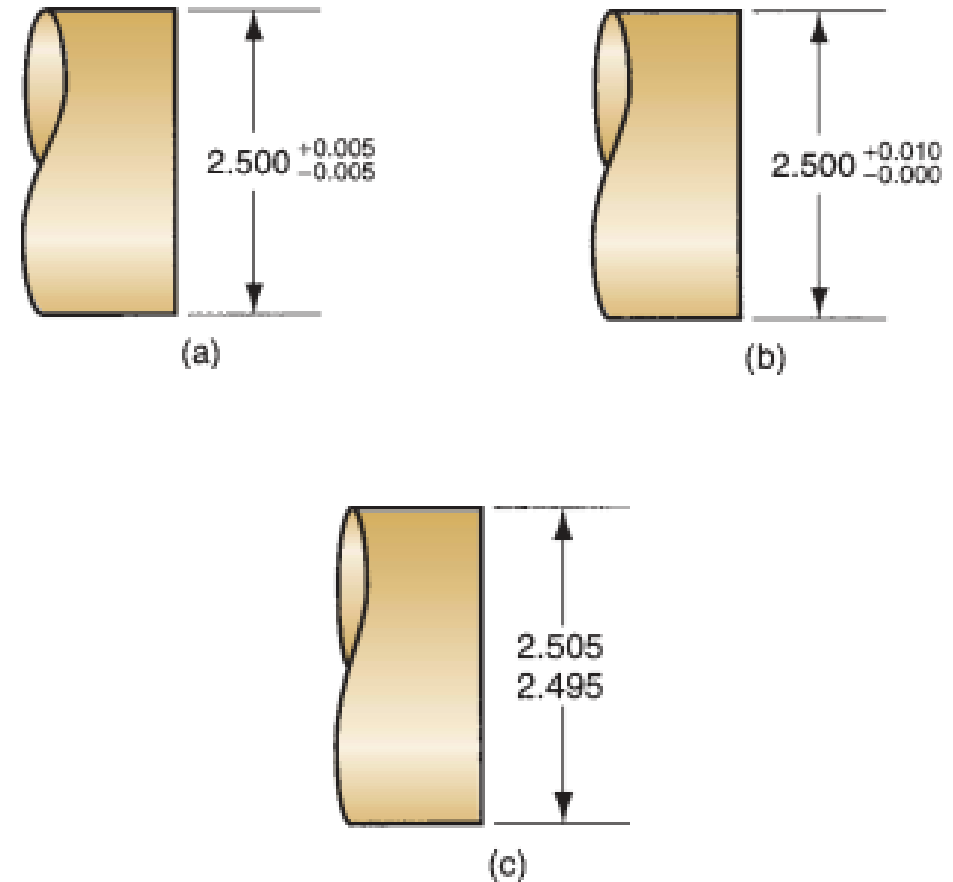


FIGURE 1: Three ways to specify tolerance limits for a nominal dimension of 2.500: (a) bilateral, (b) unilateral, and (c) limit dimensions.

Definitions of Geometric Attributes

Angularity—The extent to which a part feature such as a surface or axis is at a specified angle relative to a reference surface. If the angle = 90° , then the attribute is called perpendicularity or squareness.

Circularity—For a surface of revolution such as a cylinder, circular hole, or cone, circularity is the degree to which all points on the intersection of the surface and a plane perpendicular to the axis of revolution are equidistant from the axis. For a sphere, circularity is the degree to which all points on the intersection of the surface and a plane passing through the center are equidistant from the center.

Concentricity—The degree to which any two (or more) part features such as a cylindrical surface and a circular hole have a common axis.

Cylindricity—The degree to which all points on a surface of revolution such as a cylinder are equidistant from the axis of revolution.

Flatness—The extent to which all points on a surface lie in a single plane.

Parallelism—The degree to which all points on a part feature such as a surface, line, or axis are equidistant from a reference plane or line or axis.

Perpendicularity—The degree to which all points on a part feature such as a surface, line, or axis are 90° from a reference plane or line or axis.

Roundness—Same as circularity.

Squareness—Same as perpendicularity.

Straightness—The degree to which a part feature such as a line or axis is a straight line.

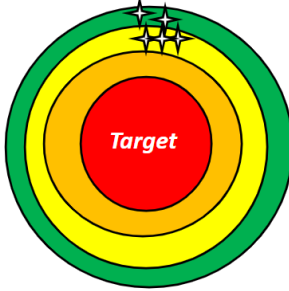
Dimension

A **dimension** is a measurable physical quantity, such as length, mass, time, or temperature, that can be expressed using a standard system of units. Dimensions provide the basis for measurement by comparing an unknown quantity with a known reference standard.

1. **U.S. Customary System (U.S.C.S.)** – used mainly in the United States.
 - Example: Length measured in **inches, feet, or miles**.
 - If we measure the dimension of a steel rod, its length might be **24 inches**.
2. **International System of Units (SI or Metric System)** – widely used internationally.
 - Example: Length measured in **millimeters, centimeters, or meters**.
 - The same steel rod dimension could be expressed as **0.61 meters (610 mm)**.



Case 1 – Low Accuracy Low Precision



Case 2 – Low Accuracy High Precision



Case 3 – High Accuracy Low Precision



Case 3 – High Accuracy High Precision

Accuracy – The degree to which the measured value agrees with the true value of the quantity of interest. A measurement is accurate when it is free from systematic errors.

Precision – The degree of repeatability in the measurement process. Good precision means random errors are minimized, and repeated measurements give consistent results.

Gaging (or Gauging) – A process that determines whether a part characteristic meets or does not meet the design specification. It is faster than measuring but provides less information about the actual value.

Precision Gage Block: Slip Gauges

- A **gauge block** (also called a gage block) is a precision standard used for calibrating and comparing dimensional measuring instruments.
- Gauge blocks are usually square or rectangular, with measuring surfaces that are finished to be **dimensionally accurate, parallel, and polished to a mirror finish**.
- This high-quality finish ensures very close contact between blocks and accuracy within a few millionths of an inch. Different grades of gauge blocks are available, offering tighter tolerances for higher precision applications.



Set of M87		
Range (mm)	Step (mm)	Pieces
1.001 to 1.009	0.001	9
1.01 to 1.49	0.01	49
0.5 to 9.5	0.5	19
10 to 90	10	9
1.005	-	1
Total		87



Measuring Instruments: Linear Dimensions or Measurements

Definition:

Linear measurement means measurement between two points or planes. (or) It is the distance between them, measured using a line or end standard.

Types of Measuring Equipment

◆ Non-Precision Instruments

- Steel Rule
- Calliper and Scale
- Feeler Gauge

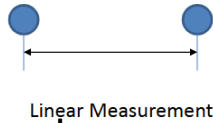
◆ Precision Instruments

- Vernier Calliper
- Vernier Height Gauge
- Vernier Depth Gauge
- Micrometre
- Inside Micrometre
- Depth Micrometre

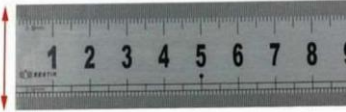
Calculating Least Count

- $L.C. = \frac{\text{Smallest Division on Main Scale}}{\text{Number of Divisions on Vernier Scale}}$
- If value of smallest division on main scale is 1mm and there are 50 numbers of divisions on vernier scale then least count will be
- $L.C. = \frac{1}{50} = 0.02mm$

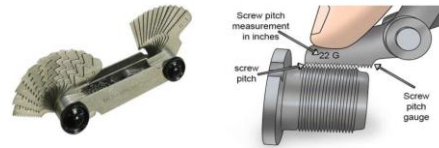
i.e. $TR = MSR + VSD \times LC$
 $TR = 18 + (23 \times 0.02) = 18 + 0.46 = 18.46$
 $TR = 18.46 \text{ mm}$



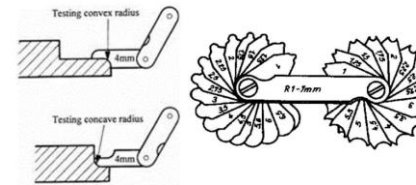
SIZE	WIDTH
150mm	18mm
200mm	20mm
300mm	25mm
500mm	28mm
600mm	30mm



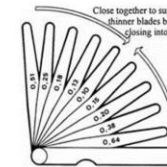
Pitch gauge



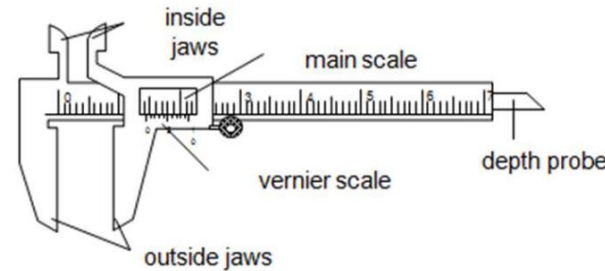
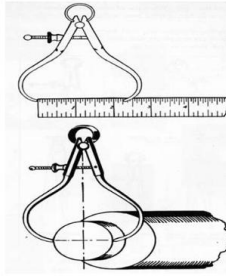
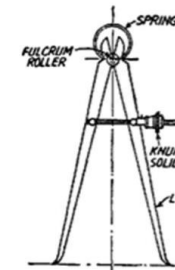
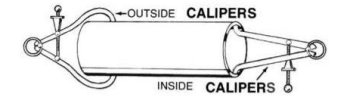
Radius Gauges



Feeler gauge



Calipers



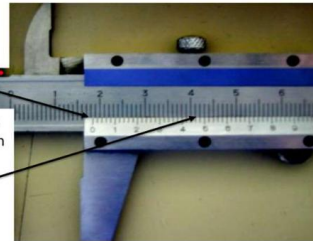
1) Read the "0" position of the vernier scale on the main scale to get a rough reading.

Main reading = 18mm

2) Look along the vernier scale until one of the vernier division coincides with the main scale

Vernier scale reading = 0.46

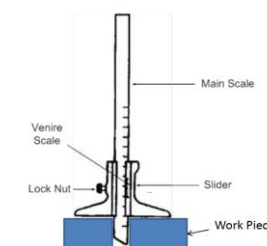
3) Accurate reading = 18 + 0.46 = 18.46mm ± 0.01 mm



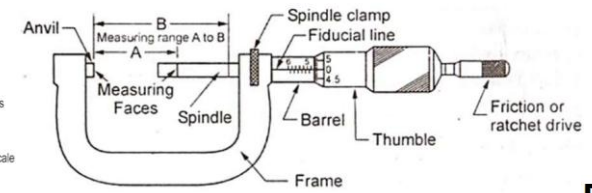
Vernier Height gauge



Vernier Depth Gauge

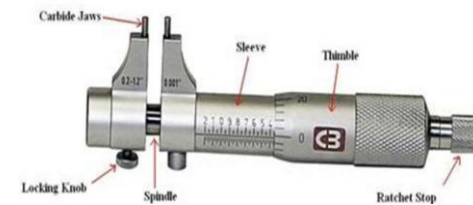


Micrometer

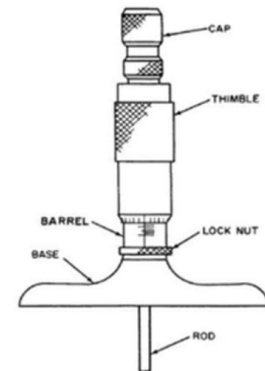


Least Count of Micrometer is = 0.01mm (i.e. 0.5/50 => Smallest division on main scale is 0.5mm and total divisions on vernier scale is 50).

Inside Micrometer



Depth Micrometer



Measuring Instruments: Comparative Instrument

Definition:

Comparative instruments are measurement tools used to make dimensional comparisons between two objects, such as a work part and a reference surface. Key characteristics include:

- They cannot provide absolute measurements of a quantity
- Instead, they measure the magnitude and direction of deviation between two objects
- They compare one object against another reference object or surface
- Examples include both mechanical and electronic gauges

1. Dial Gauge (Dial Indicator)

A dial gauge, more specifically called a **dial indicator**, is the most common type of mechanical comparative instrument. Its key features are:

Function: Converts and amplifies linear movement of a contact pointer into rotation of a dial needle



2. Electronic Gauges

Electronic gages are a family of measuring and gaging instruments based on transducers that convert linear displacement into electrical signals. The electrical signal is then amplified and transformed into suitable data formats such as digital readouts.



Measuring Instruments: Fixed Guages

Definition:

Fixed Gage A physical replica of the part dimension to be assessed. Used to directly check whether a part meets the required size or tolerance without reading measurements.

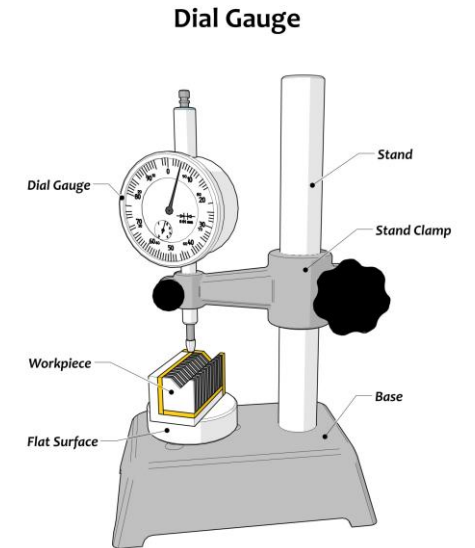
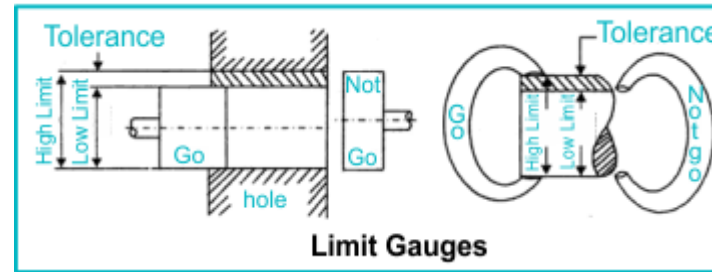
1. Master Gage

- A direct replica of the nominal size of the part dimension.
- Used mainly for setting up comparative measuring instruments (e.g., dial indicators) or for calibrating other measuring devices.

2. Limit Gage: A reverse replica of the part dimension, designed to check dimensions at one or more tolerance limits.

Typically consists of two gauges in one piece:

- One checks the **lower limit** of tolerance.
- The other checks the **upper limit** of tolerance.



3. GO/NO-GO Gages

GO Limit:

- Allows the part to pass through or be inserted.
- Checks the maximum material condition (smallest hole size or largest shaft size).

NO-GO Limit:

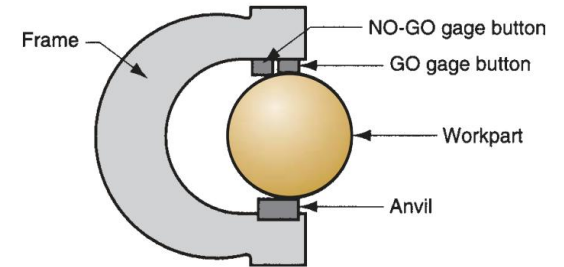
- Prevents the part from passing through.
- Checks the minimum material condition (largest hole size or smallest shaft size).



Measuring Instruments: Fixed Guages

4. Snap Gauge

- A type of limit gage used for checking outside part dimensions.
- Usually has a C-shaped frame with gaging surfaces in the jaws.



5. Ring Gauge: A ring gage is a fixed gage used for checking cylindrical diameters of parts.

- It is a ring whose opening is machined to one of the tolerance limits of the part diameter.

Typically used in pairs:

- GO Ring Gauge – checks the maximum material condition (largest acceptable shaft).
- NO-GO Ring Gauge – checks the minimum material condition.
- For easy handling, the outside of the ring is often knurled, and the NO-GO ring is usually marked with a groove.

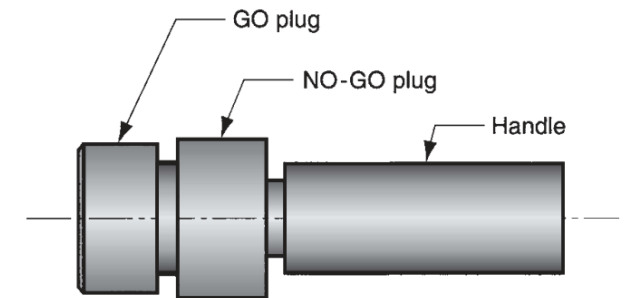


6. Plug Gauge: A plug gauge is a fixed gauge used for checking hole diameters.

- It typically consists of a handle with two accurately ground cylindrical plugs of hardened steel.
- One plug serves as the GO gage (checks lower limit), and the other as the NO-GO gage (checks upper limit).

Variants include:

- Taper Gages – for checking tapered holes.
- Thread Gages – for checking internal threads.



Thank You...